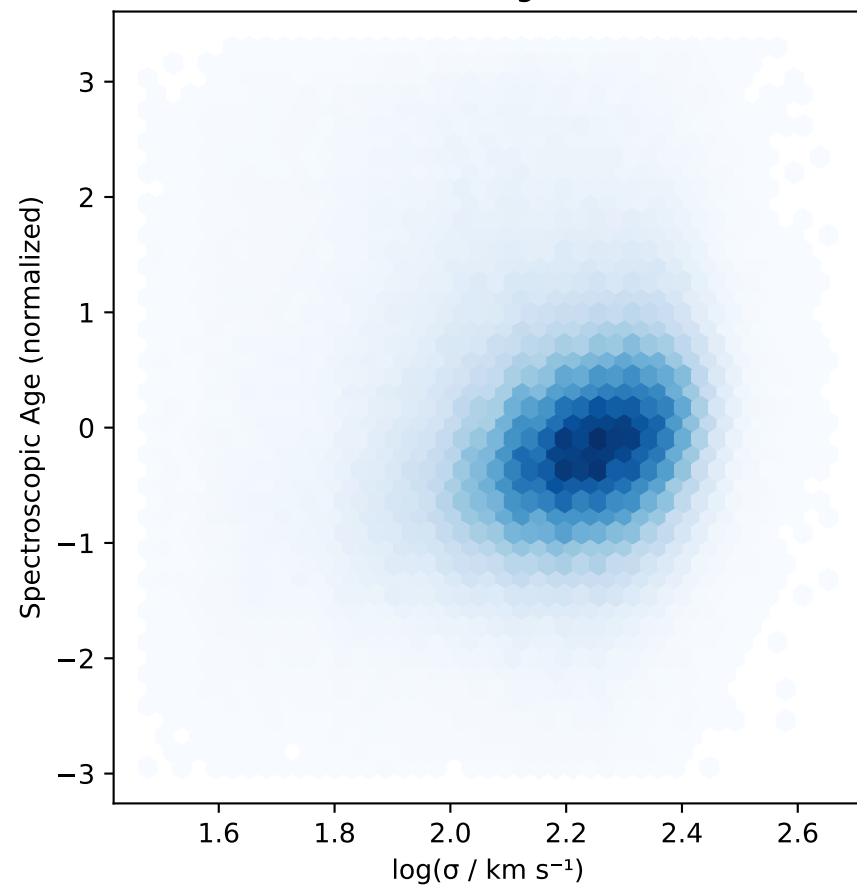
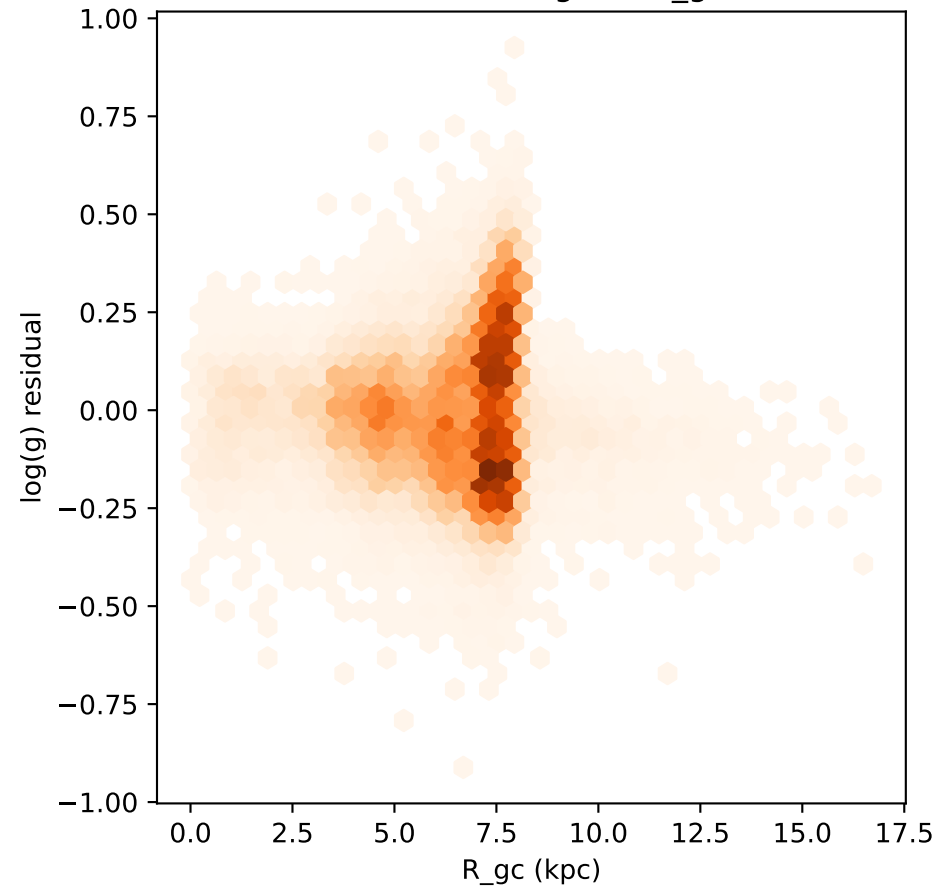
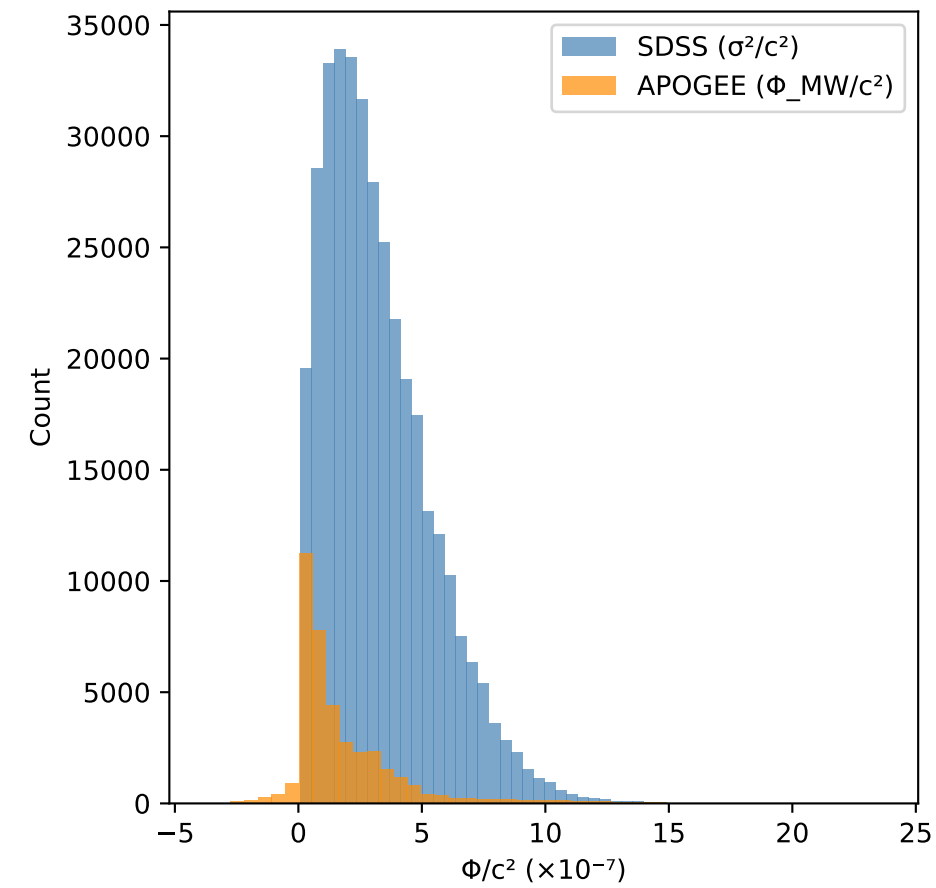
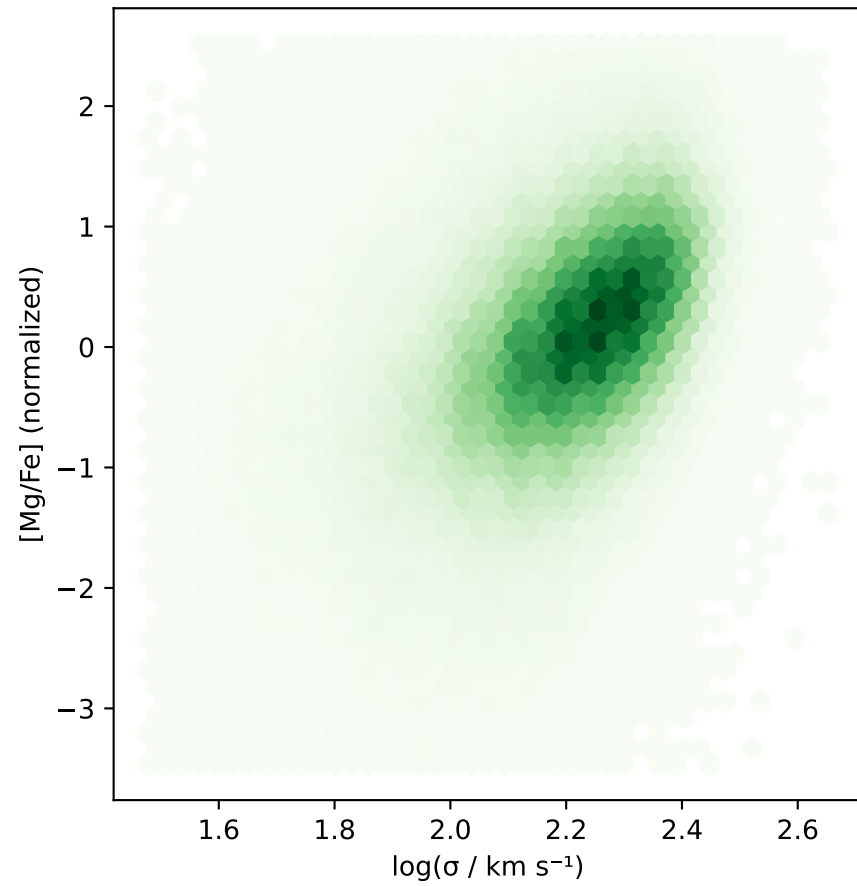
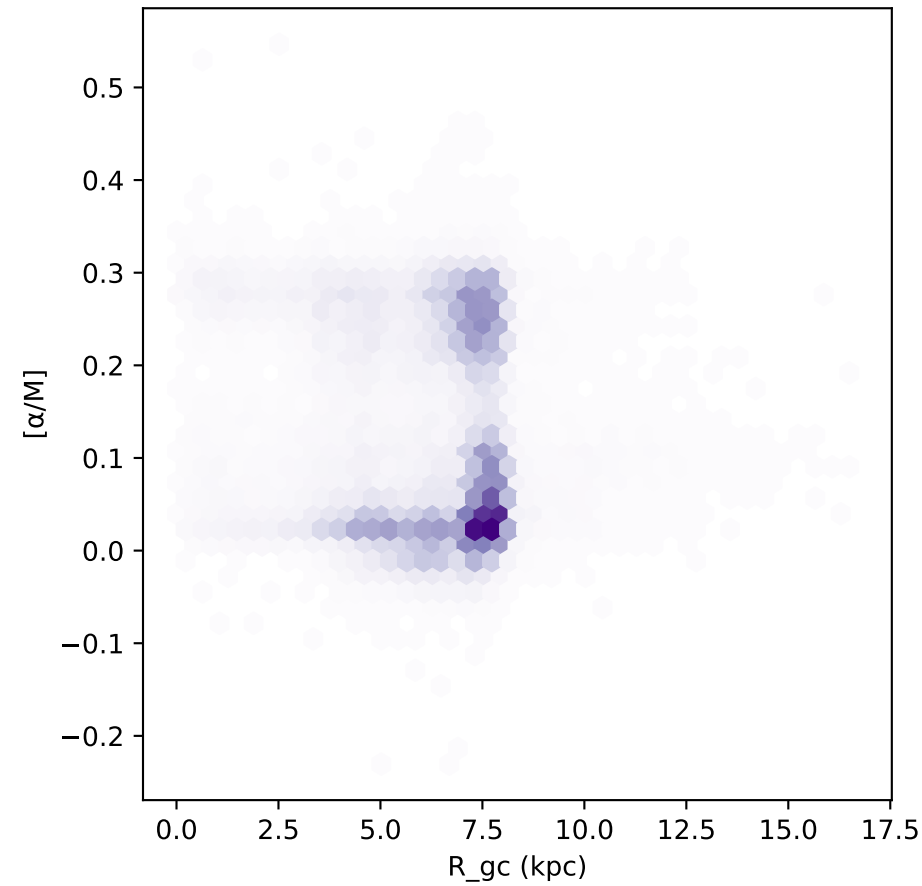


A. SDSS: Age vs σ B. APOGEE: Age vs R_{gc} 

C. Potential Distribution

D. SDSS: $[Mg/Fe]$ vs σ E. APOGEE: $[\alpha/M]$ vs R_{gc} 

SDSS vs APOGEE COMPARISON

SDSS (361k galaxies):

Age vs σ : $r = -0.024$ (younger at higher σ) $[Mg/Fe]$ vs σ : $r = +0.258$

Discrepancy: TEP-consistent

APOGEE (40k stars):

Age vs R_{gc} : $r = +0.064$ (older at outer R) $[\alpha/M]$ vs R_{gc} : $r = -0.029$

Discrepancy: TEP-inconsistent

KEY DIFFERENCES:

1. Potential: σ^2 vs Galactic Φ
2. Age: D4000/H β vs $\log(g)$
3. Scale: Extragalactic vs Galactic

CONCLUSION:

Results not directly comparable.
 Different physical regimes probed.
 SDSS result requires caution.